

Econ 326
Assignment 6 — Solutions

1. **Regression of liver on alcohol.**

From the output we have: $\hat{\beta}_1 = 3.5864$, $SE(\hat{\beta}_1) = 0.7541$, and $df = n - 2 = 19$.

(a) **Find A–D.**

A (t value): The t -statistic is the ratio of the estimate to its standard error:

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)} = \frac{3.5864}{0.7541} = 4.756.$$

Therefore **A = 4.756**.

B (p-value): The p-value for a two-sided test of $H_0: \beta_1 = 0$ is

$$p = 2 \cdot P(t_{19} > |t|) = 2 \cdot P(t_{19} > 4.756).$$

In R: `2 * pt(-4.756, df = 19) ≈ 0.0001`.

Therefore **B ≈ 0.0001**.

C and D (95% confidence interval): The 95% confidence interval for β_1 is

$$\hat{\beta}_1 \pm t_{0.025,19} \cdot SE(\hat{\beta}_1),$$

where $t_{0.025,19} = 2.093$ (in R: `qt(0.975, df = 19)`). Substituting:

$$\begin{aligned} \text{Lower bound (C)} &= 3.5864 - 2.093 \times 0.7541 \\ &= 3.5864 - 1.5783 \\ &= 2.008. \end{aligned}$$

$$\begin{aligned} \text{Upper bound (D)} &= 3.5864 + 2.093 \times 0.7541 \\ &= 3.5864 + 1.5783 \\ &= 5.165. \end{aligned}$$

Therefore **C = 2.008** and **D = 5.165**.

(b) **Test $H_0: \beta_1 = 5$ vs. $H_1: \beta_1 \neq 5$ at 5%.**

Approach 1: t-test.

The test statistic is

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)} = \frac{3.5864 - 5}{0.7541} = \frac{-1.4136}{0.7541} = -1.875.$$

The decision rule at 5% significance (two-sided) is: reject H_0 if $|t| > t_{0.025,19} = 2.093$ (in R: `qt(0.975, df = 19)`).

Since $|t| = 1.875 < 2.093$, we **fail to reject** H_0 at the 5% significance level.

Approach 2: Confidence interval.

From part (a), the 95% confidence interval for β_1 is $[2.008, 5.165]$. Since the hypothesized value $\beta_1 = 5$ falls inside this interval, we **fail to reject** H_0 at the 5% significance level.

(c) **Test the same hypothesis at 10%.**

Approach 1: t-test.

The test statistic is the same as in part (b): $t = -1.875$. The decision rule at 10% significance (two-sided) is: reject H_0 if $|t| > t_{0.05,19} = 1.729$ (in R: `qt(0.95, df = 19)`).

Since $|t| = 1.875 > 1.729$, we **reject** H_0 at the 10% significance level.

Approach 2: Confidence interval.

The 90% confidence interval for β_1 is

$$\hat{\beta}_1 \pm t_{0.05,19} \cdot \text{SE}(\hat{\beta}_1) = 3.5864 \pm 1.729 \times 0.7541 = 3.5864 \pm 1.3034 = [2.283, 4.890],$$

where $t_{0.05,19} = 1.729$ is obtained in R via `qt(0.95, df = 19)`. Since the hypothesized value $\beta_1 = 5$ falls outside this interval ($5 > 4.890$), we **reject** H_0 at the 10% significance level.

2. Regression of `ln_heart` on `ln_alcohol`.

From the output we have: $n = 21$, $\text{df} = n - 2 = 19$, and the 95% confidence interval for the coefficient on `ln_alcohol` is $[-0.6114, -0.0952]$.

(a) **Find A–D.**

A (Estimate): The point estimate is the midpoint of the 95% confidence interval:

$$\hat{\beta}_1 = \frac{-0.6114 + (-0.0952)}{2} = \frac{-0.7066}{2} = -0.3533.$$

Therefore **A** = **-0.3533**.

B (Standard Error): The 95% confidence interval has the form

$$\hat{\beta}_1 \pm t_{0.025,19} \cdot \text{SE}(\hat{\beta}_1),$$

so the half-width of the interval equals $t_{0.025,19} \cdot \text{SE}(\hat{\beta}_1)$. The half-width is

$$\frac{-0.0952 - (-0.6114)}{2} = \frac{0.5162}{2} = 0.2581.$$

Since $t_{0.025,19} = 2.093$ (in R: `qt(0.975, df = 19)`), we solve for the standard error:

$$\text{SE}(\hat{\beta}_1) = \frac{0.2581}{2.093} = 0.1233.$$

Therefore **B = 0.1233**.

C (t value): The t -statistic is

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{-0.3533}{0.1233} = -2.865.$$

Therefore **C = -2.865**.

D (p-value): The p-value for a two-sided test of $H_0: \beta_1 = 0$ is

$$p = 2 \cdot P(t_{19} < -2.865).$$

In R: `2 * pt(-2.865, df = 19) ≈ 0.010`.

Therefore **D ≈ 0.010**.

(b) **Interpretation of the coefficient on `ln_alcohol`.**

This is a log-log regression model:

$$\ln(\text{heart}) = \beta_0 + \beta_1 \ln(\text{alcohol}) + u.$$

In a log-log model, the slope coefficient β_1 represents the **elasticity** of the dependent variable with respect to the independent variable. Specifically, β_1 measures the percentage change in heart disease deaths associated with a 1% change in alcohol consumption.

The estimated coefficient is $\hat{\beta}_1 = -0.3533$. This means that a 1% increase in alcohol consumption is associated with approximately a 0.353% **decrease** in heart disease deaths per 100,000 people.

(c) **Test $H_0: \beta_1 \geq 0$ vs. $H_1: \beta_1 < 0$ at 5%.**

This is a one-sided (left-tailed) test. The test statistic is

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{-0.3533}{0.1233} = -2.865.$$

The decision rule at 5% significance (one-sided, left tail) is: reject H_0 if $t < -t_{0.05,19}$, where $t_{0.05,19} = 1.729$ (in R: `qt(0.95, df = 19)`).

Since $t = -2.865 < -1.729$, we **reject** H_0 at the 5% significance level. There is statistically significant evidence that the coefficient on `ln_alcohol` is negative.

3. Regression of y on x .

From the output we have: $\hat{\beta}_1 = 0.3049$, $\text{SE}(\hat{\beta}_1) = 0.1774$, and $\text{df} = 23$.

(a) **Find A–D.**

A (t value): The t -statistic is

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{0.3049}{0.1774} = 1.719.$$

Therefore **A = 1.719**.

B (p-value): The p-value for a two-sided test of $H_0: \beta_1 = 0$ is

$$p = 2 \cdot P(t_{23} > |t|) = 2 \cdot P(t_{23} > 1.719).$$

In R: `2 * pt(-1.719, df = 23) ≈ 0.099`.

Therefore **B ≈ 0.099**.

C and D (95% confidence interval): The 95% confidence interval for β_1 is

$$\hat{\beta}_1 \pm t_{0.025,23} \cdot \text{SE}(\hat{\beta}_1),$$

where $t_{0.025,23} = 2.069$ (in R: `qt(0.975, df = 23)`). Substituting:

$$\begin{aligned} \text{Lower bound (C)} &= 0.3049 - 2.069 \times 0.1774 \\ &= 0.3049 - 0.3670 \\ &= -0.062. \end{aligned}$$

$$\begin{aligned} \text{Upper bound (D)} &= 0.3049 + 2.069 \times 0.1774 \\ &= 0.3049 + 0.3670 \\ &= 0.672. \end{aligned}$$

Therefore **C = -0.062** and **D = 0.672**.

(b) **Test $H_0: \beta_1 = 0$ vs. $H_1: \beta_1 \neq 0$ at 5%.**

The test statistic is

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{0.3049}{0.1774} = 1.719.$$

The decision rule at 5% significance (two-sided) is: reject H_0 if $|t| > t_{0.025,23} = 2.069$ (in R: `qt(0.975, df = 23)`).

Since $|t| = 1.719 < 2.069$, we **fail to reject** H_0 at the 5% significance level.

Alternatively, from the 95% confidence interval $[-0.062, 0.672]$, we see that 0 falls inside the interval, which confirms that we fail to reject H_0 at 5%.

(c) **Test $H_0: \beta_1 \leq 0$ vs. $H_1: \beta_1 > 0$ at 5%.**

This is a one-sided (right-tailed) test. The test statistic is

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} = \frac{0.3049}{0.1774} = 1.719.$$

The decision rule at 5% significance (one-sided, right tail) is: reject H_0 if $t > t_{0.05,23}$, where $t_{0.05,23} = 1.714$ (in R: `qt(0.95, df = 23)`).

Since $t = 1.719 > 1.714$, we **reject** H_0 at the 5% significance level. There is statistically significant evidence that $\beta_1 > 0$.

(d) **Explanation of the difference between parts (b) and (c).**

The two-sided test in part (b) fails to reject, while the one-sided test in part (c) rejects. This apparent contradiction arises because the two tests use different critical values:

- The **two-sided test** splits the 5% significance level across both tails of the t -distribution, placing 2.5% in each tail. This yields a critical value of $t_{0.025,23} = 2.069$ (`qt(0.975, df = 23)`). The test rejects when $|t| > 2.069$.
- The **one-sided test** places the entire 5% in one tail (the right tail, since $H_1: \beta_1 > 0$). This yields a smaller critical value of $t_{0.05,23} = 1.714$ (In R: `qt(0.95, df = 23)`). The test rejects when $t > 1.714$.

Since the observed t -statistic is 1.719, it satisfies $1.714 < 1.719 < 2.069$, falling in the region where the one-sided test rejects but the two-sided test does not.

The one-sided test has more **power** against the specific alternative $\beta_1 > 0$ because it concentrates the entire rejection region in the direction of interest. It does not need to guard against the possibility that $\beta_1 < 0$. However, this comes at the cost of being unable to detect effects in the opposite direction.

4. Power of one-sided and two-sided tests.

We test $H_0: \beta = 1$ at $\alpha = 0.05$ with known variance $\text{Var}(\hat{\beta}) = 4$, so $\sigma_{\hat{\beta}} = 2$. Since the variance is known, the test statistic

$$Z = \frac{\hat{\beta} - 1}{\sigma_{\hat{\beta}}} = \frac{\hat{\beta} - 1}{2}$$

follows a standard normal distribution under H_0 .

(a) **Rejection regions.**

Two-sided test ($H_1: \beta \neq 1$): Reject H_0 if $|Z| > z_{0.025} = 1.96$. In terms of $\hat{\beta}$:

$$\begin{aligned} \left| \frac{\hat{\beta} - 1}{2} \right| > 1.96 &\iff |\hat{\beta} - 1| > 3.92 \\ &\iff \hat{\beta} > 4.92 \text{ or } \hat{\beta} < -2.92. \end{aligned}$$

One-sided test ($H_1: \beta > 1$): Reject H_0 if $Z > z_{0.05} = 1.645$. In terms of $\hat{\beta}$:

$$\frac{\hat{\beta} - 1}{2} > 1.645 \iff \hat{\beta} > 1 + 2 \times 1.645 = 4.29.$$

(b) **Power of the two-sided test.**

Under the true value β^* , we have $\hat{\beta} \sim N(\beta^*, 4)$, so $Z = (\hat{\beta} - 1)/2 \sim N(\delta, 1)$ where $\delta = (\beta^* - 1)/2$. The power of the two-sided test is

$$\begin{aligned}\text{Power} &= P(\hat{\beta} > 4.92) + P(\hat{\beta} < -2.92) \\ &= P(Z > 1.96 - \delta) + P(Z < -1.96 - \delta),\end{aligned}$$

where $Z \sim N(0, 1)$.

Note that the power depends on β^* only through $|\delta|$, since replacing δ with $-\delta$ simply swaps the two tail probabilities. Therefore the power is the same for values of β^* that are equidistant from $\beta_0 = 1$. It suffices to compute two cases.

- $\beta^* = -3$: $\delta = (-3 - 1)/2 = -2$.

$$\begin{aligned}\text{Power} &= P(Z > 1.96 - (-2)) + P(Z < -1.96 - (-2)) \\ &= P(Z > 3.96) + P(Z < 0.04) = 0.000 + 0.516 = 0.516.\end{aligned}$$

- $\beta^* = 0$: $\delta = (0 - 1)/2 = -0.5$.

$$\begin{aligned}\text{Power} &= P(Z > 1.96 + 0.5) + P(Z < -1.96 + 0.5) \\ &= P(Z > 2.46) + P(Z < -1.46) = 0.007 + 0.072 = 0.079.\end{aligned}$$

- $\beta^* = 2$: By symmetry (same distance from $\beta_0 = 1$ as $\beta^* = 0$), Power = 0.079.
- $\beta^* = 5$: By symmetry (same distance from $\beta_0 = 1$ as $\beta^* = -3$), Power = 0.516.

(c) **Power of the one-sided test.**

The one-sided test rejects when $\hat{\beta} > 4.29$. The power is

$$\text{Power} = P(\hat{\beta} > 4.29) = P(Z > 1.645 - \delta),$$

where $Z \sim N(0, 1)$ and $\delta = (\beta^* - 1)/2$.

- $\beta^* = -3$: $\delta = -2$.

$$\text{Power} = P(Z > 1.645 + 2) = P(Z > 3.645) \approx 0.0001.$$

- $\beta^* = 0$: $\delta = -0.5$.

$$\text{Power} = P(Z > 1.645 + 0.5) = P(Z > 2.145) = 0.016.$$

- $\beta^* = 2$: $\delta = 0.5$.

$$\text{Power} = P(Z > 1.645 - 0.5) = P(Z > 1.145) = 0.126.$$

- $\beta^* = 5$: $\delta = 2$.

$$\text{Power} = P(Z > 1.645 - 2) = P(Z > -0.355) = 0.639.$$

(d) **Comparison.**

Summarizing the results:

β^*	δ	Two-sided power	One-sided power
-3	-2	0.516	0.0001
0	-0.5	0.079	0.016
2	0.5	0.079	0.126
5	2	0.516	0.639

Key observations:

1. The two-sided test has **symmetric power** around $\beta_0 = 1$: the power at $\beta^* = -3$ (distance 4 below 1) equals the power at $\beta^* = 5$ (distance 4 above 1), and similarly the power at $\beta^* = 0$ equals the power at $\beta^* = 2$ (both distance 1 from 1).
2. The one-sided test has **higher power** than the two-sided test when $\beta^* > 1$, i.e., when the true value lies in the direction of the alternative hypothesis $H_1: \beta > 1$. For example, at $\beta^* = 5$, the one-sided power is 0.639 compared to 0.516 for the two-sided test.
3. The one-sided test has **near-zero power** when $\beta^* < 1$. For instance, at $\beta^* = -3$, the one-sided power is only 0.0001, meaning the test almost never rejects even though β^* is far from $\beta_0 = 1$.
4. This illustrates the fundamental **trade-off**: the one-sided test concentrates its entire rejection region in the direction of H_1 , which gives it more power to detect departures in that direction. However, it sacrifices almost all ability to detect departures in the opposite direction. The two-sided test distributes its rejection region across both tails, making it less powerful in any single direction but able to detect departures on either side.

5. **Minimum detectable effect.**

We test $H_0: \beta = 0$ vs. $H_1: \beta \neq 0$ at $\alpha = 0.05$ with known variance. For $\beta^* > 0$, the power is approximately

$$\text{Power} \approx 1 - \Phi\left(1.96 - \frac{\beta^*}{\sigma_{\hat{\beta}}}\right).$$

We want the smallest $\beta^* > 0$ such that $\text{Power} \geq 0.80$.

(a) **MDE when $\sigma_{\hat{\beta}} = 3$.**

Set Power = 0.80 and solve:

$$1 - \Phi\left(1.96 - \frac{\beta^*}{3}\right) = 0.80$$

$$\Phi\left(1.96 - \frac{\beta^*}{3}\right) = 0.20$$

$$1.96 - \frac{\beta^*}{3} = \Phi^{-1}(0.20) = -0.84$$

$$\frac{\beta^*}{3} = 1.96 + 0.84 = 2.80$$

$$\beta^* = 8.40.$$

By symmetry, the power is also 80% when $\beta^* = -8.40$. Therefore **MDE = 8.40**.

(b) **MDE when $\sigma_{\hat{\beta}} = 1.5$.**

The same steps give:

$$1.96 - \frac{\beta^*}{1.5} = -0.84$$

$$\frac{\beta^*}{1.5} = 2.80$$

$$\beta^* = 4.20.$$

Therefore **MDE = 4.20**.

Note that halving $\sigma_{\hat{\beta}}$ (from 3 to 1.5) also halved the MDE (from 8.40 to 4.20).

(c) **General formula and the role of sample size.**

In general, the equation $1.96 - \beta^*/\sigma_{\hat{\beta}} = -0.84$ gives

$$\text{MDE} = (z_{0.975} + z_{0.80}) \sigma_{\hat{\beta}} = (1.96 + 0.84) \sigma_{\hat{\beta}} = 2.80 \sigma_{\hat{\beta}}.$$

The MDE is **directly proportional** to $\sigma_{\hat{\beta}}$. Since $\sigma_{\hat{\beta}}$ typically decreases with the sample size, a larger sample reduces the MDE, allowing the test to detect smaller effects. Conversely, a small sample implies a large MDE, meaning that only large effects are likely to be detected.

6. Distribution of the p-value under H_0 .

(a) **Show that $F(X) \sim \text{Uniform}(0, 1)$.**

Let $U = F(X)$. We need to show $P(U \leq u) = u$ for all $u \in (0, 1)$.

Since F is strictly increasing, F^{-1} exists and $F(x) \leq u \iff x \leq F^{-1}(u)$.

Therefore:

$$\begin{aligned} P(U \leq u) &= P(F(X) \leq u) \\ &= P(X \leq F^{-1}(u)) \\ &= F(F^{-1}(u)) \\ &= u. \end{aligned}$$

The third equality uses the definition of the CDF ($P(X \leq x) = F(x)$) and the fourth uses the fact that F and F^{-1} are inverses. Since $P(U \leq u) = u$ for all $u \in (0, 1)$, we conclude $U \sim \text{Uniform}(0, 1)$.

(b) **Show that $V = 2(1 - \Phi(|Z|)) \sim \text{Uniform}(0, 1)$.**

We compute $P(V \leq v)$ directly for $v \in (0, 1)$:

$$\begin{aligned} P(V \leq v) &= P(2(1 - \Phi(|Z|)) \leq v) \\ &= P(\Phi(|Z|) \geq 1 - v/2) \\ &= P(|Z| \geq \Phi^{-1}(1 - v/2)) \\ &= P(|Z| \geq z_{1-v/2}), \end{aligned}$$

where the third line uses the fact that Φ is strictly increasing, so $\Phi(|Z|) \geq 1 - v/2 \iff |Z| \geq \Phi^{-1}(1 - v/2)$. The quantity $z_{1-v/2}$ is exactly the critical value of a two-sided test at significance level v .

Since the standard normal distribution is symmetric around zero, $P(Z \geq c) = P(Z \leq -c)$ for any c , so

$$P(|Z| \geq z_{1-v/2}) = P(Z \geq z_{1-v/2}) + P(Z \leq -z_{1-v/2}) = 2P(Z \geq z_{1-v/2}).$$

By definition, $P(Z \leq z_{1-v/2}) = 1 - v/2$, hence $P(Z \geq z_{1-v/2}) = v/2$ and

$$P(|Z| \geq z_{1-v/2}) = 2 \cdot v/2 = v.$$

Since $P(V \leq v) = v$ for all $v \in (0, 1)$, we conclude $V \sim \text{Uniform}(0, 1)$.

(c) **The p-value is uniform under H_0 .**

Under H_0 , the test statistic $Z = (\hat{\beta} - \beta_0)/\sigma_{\hat{\beta}}$ has a standard normal distribution. The p-value is $p = 2(1 - \Phi(|Z|))$, which is exactly the random variable V from Part (b) with $Z \sim N(0, 1)$. By Part (b), $p \sim \text{Uniform}(0, 1)$ under H_0 .

(d) **The size of the test equals α .**

The test rejects H_0 when $p < \alpha$. The size is

$$P(\text{reject } H_0 \mid H_0) = P(p < \alpha \mid H_0).$$

Under H_0 , $p \sim \text{Uniform}(0, 1)$ by Part (c). Since the $\text{Uniform}(0, 1)$ distribution is continuous,

$$P(p < \alpha) = P(p \leq \alpha) = \alpha,$$

where the last equality uses the CDF of the $\text{Uniform}(0, 1)$ distribution. Therefore the size of the test is α .